ES301 – Energy & Environmental Engineering

Unit – I (Introduction to Energy Science)

- **Main**
  - Energy Systems
  - Energy Resources
  - Sustainability
- **Detail**
  - Energy systems and resources
  - Energy – sustainability – environment
  - Energy sources – transformations – efficiency – storage
  - Fossil fuels (coal – oil – shale – sands – gasification)
  - Alternatives (biomass – wind – solar – nuclear – tidal – hydrogen)
  - Environmental trade-offs
  - Energy storage (pumped hydro – superconductors – batteries)

Unit – II (Ecosystems)

- **Main**
  - Ecosystem Concepts
  - Structure & Function
  - Types of Ecosystems
- **Detail**
  - Producers – consumers – decomposers
  - Energy flow – ecological succession
  - Food chain – food web – ecological pyramids
  - Forest – grassland – desert ecosystems
  - Aquatic ecosystems (pond – lake – river – ocean – estuary)

Unit – III (Biodiversity & Conservation)

- **Main**
  - Biodiversity

    - Threats
    - Conservation
- **Detail**
    - Genetic – species – ecosystem diversity
    - Bio-geographical classification of India
    - Values of biodiversity
    - India as mega diversity nation
    - Hotspots of biodiversity
    - Threats (habitat loss – poaching – man-wildlife conflict)
    - In-situ – Ex-situ conservation

Unit – IV (Environmental Pollution)

- **Main**
    - Pollution Types
    - Solid Waste
    - Disaster Management
- **Detail**
    - Air – water – soil – marine – noise – thermal pollution
    - Nuclear hazards
    - Solid waste management
    - Role of individual
    - Pollution case studies
    - Flood – earthquake – cyclone – landslide

Unit – V (Social Issues & Environment)

- **Main**
    - Sustainable Development
    - Environmental Ethics
    - Environmental Laws
- **Detail**
    - Sustainable development
    - Urban energy problems
    - Water conservation – rain water harvesting – watershed

- Rehabilitation issues
- Climate change – global warming – acid rain – ozone depletion
- Environmental protection acts
- Public awareness

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302 – Discrete Structure

Unit – I

- **Main**
  - Set Theory
  - Relations
  - Functions
- **Detail**
  - Types of sets – Venn diagrams
  - Relations – equivalence – partial order
  - Functions – one to one – onto – inverse
  - Pigeonhole principle

Unit – II

- **Main**
  - Propositional Logic
  - Predicate Logic
  - Normal Forms
- **Detail**
  - Truth tables – tautology – contradiction
  - Logical equivalence – implication
  - Predicates – quantifiers
  - Normal forms

Unit – III

- **Main**
  - Algebraic Structures
  - Groups
  - Rings & Fields
- **Detail**
  - Semigroup – monoid – group
  - Abelian group – cyclic group

    - Subgroup – cosets – homomorphism
    - Rings – fields

Unit – IV

- **Main**
    - Graph Theory
    - Paths & Cycles
    - Graph Coloring
- **Detail**
    - Graph terminology
    - Euler – Hamiltonian paths
    - Shortest path
    - Graph coloring – chromatic number

Unit – V

- **Main**
    - Combinatorics
    - Recurrence Relations
    - Generating Functions
- **Detail**
    - Permutation – combination
    - Binomial theorem
    - Linear recurrence relations
    - Generating functions

CS303 – Data Structure

Unit – I

- **Main**
  - Data Structures Basics
  - Arrays
  - Linked Lists
- **Detail**
  - ADT – memory representation
  - Array operations
  - Singly – doubly – circular linked lists
  - Polynomial manipulation

Unit – II

- **Main**
  - Stack
  - Queue
  - Applications
- **Detail**
  - Stack operations – recursion
  - Infix to postfix
  - Queue – circular queue
  - Deque – priority queue

Unit – III

- **Main**
  - Trees
  - BST
  - Heap
- **Detail**
  - Tree terminology
  - BST operations

    - AVL tree
    - B-tree – B+ tree – Red black tree

Unit – IV

- **Main**
    - Graphs
    - Traversal
    - Algorithms
- **Detail**
    - Graph representation
    - DFS – BFS
    - Kruskal – Prim
    - Dijkstra algorithm

Unit – V

- **Main**
    - Sorting
    - Searching
    - Hashing
- **Detail**
    - Bubble – quick – merge – heap sort
    - Sequential – binary search
    - Hashing – indexing

CS304 – Digital Systems

Unit – I

- **Main**
  - Number Systems
  - Boolean Algebra
  - Logic Gates
- **Detail**
  - Base conversion
  - Boolean functions
  - K-map – SOP – POS
  - NAND – NOR implementation

Unit – II

- **Main**
  - Combinational Circuits
  - Arithmetic Circuits
  - MUX / DEMUX
- **Detail**
  - Half – full adder
  - Encoder – decoder
  - ALU

Unit – III

- **Main**
  - Sequential Circuits
  - Flip Flops
  - Memories
- **Detail**
  - SR – JK – D – T flip flops
  - Counters – shift registers
  - ROM – PLA – DRAM – FLASH

Unit – IV

- **Main**
  - Converters
  - Multivibrators
  - Logic Families
- **Detail**
  - A/D – D/A convertors
  - IC 555
  - TTL – CMOS – NMOS

Unit – V

- **Main**
  - Digital Communication
  - Modulation
  - Theorems
- **Detail**
  - Nyquist theorem
  - PCM – TDM
  - BPSK – BFSK
  - Shannon theorem

CS305 – Object Oriented Programming & Methodology

Unit – I

- **Main**
  - OOP Concepts
  - OO Paradigm
  - Comparison
- **Detail**
  - Procedural vs OOP
  - Object model
  - Merits – demerits

Unit – II

- **Main**
  - Encapsulation
  - Classes
  - Objects
- **Detail**
  - Data abstraction
  - Attributes – methods
  - Constructors – destructors

Unit – III

- **Main**
  - Inheritance
  - Association
  - Interfaces
- **Detail**
  - Types of inheritance
  - Aggregation
  - Abstract classes

Unit – IV

- **Main**
  - Polymorphism
  - Overloading
  - Overriding
- **Detail**
  - Compile time – runtime polymorphism

Unit – V

- **Main**
  - Strings
  - Exception Handling
  - Multithreading
- **Detail**
  - String handling
  - Exception handling
  - Data collections
  - Case studies (ATM – Library system